CELLO

Agency Prepayment: Refinance

May 2016

1 Modeling Refinances – A Three Step Process

- In order to obtain Refinances from total prepayments, we need to subtract the Turnover, Default and Curtailment forecasts produced by the model:
- Use a three step process to fit the multipliers for the Refinancing sub-model:
  - Use pre-2007 no-Media Effect data to build a baseline S-curve;
  - Use pre-2007 data to fit the Media Effect as a multiplier to the baseline S-Curve
  - Use the entire data set to fit all other multipliers (Credit Availability, WALA, burnout, loan size, CLTV etc.)

2 Pool Level Data Set

- Consistent with our overall approach, we use pool level information to fit and track the refinance module.
- The following pool level information is calculated using weighted-average loan level data:
  - Incentive, Burnout, Refi-Eligibility%, HARPed%

3 Data Updates and Model Extensions

- Incentive: Incorporated cross-sector incentive (Conventional to FHA)
- Burnout: Included LLPA and PMI when calculating relative incentive
- Refi-Eligible %: Based on LLPA and PMI underwriting guidelines
- Origination Channel: Using pool-level info on third party originations

Refinance model v1

3 Step Fitting Process

Use three cuts of the pool-level data set

7 Parameters

S Curve
Media Effect
Credit Availability
WALA
Burnout
Loan Size
CLTV

Updates

1. Incorporate crossover incentive

High LTV (>90%) conventional borrowers also have the option of refinancing into an FHA mortgage.

2. Update Burnout with LLPA and PMI

When calculating the relative incentive, incorporate LLPA and PMI. Also, the calculation is performed at the loan-level.

3. Add Refi Eligibility flag

Based on LLPA and PMI underwriting guidelines.

4. Add HARPed %

New parameter that measures the pool % that is HARPed. These borrowers tend to refi slower.

5. Add new CLTV curve for HARP eligible pools

Have two CLTV curves for normal mortgages and HARP eligible mortgages.

6. Add constraints for extrapolation

Using cap and floors for out of range extrapolation. Make sure the tail performance is reasonable.

Refinance model v1

3 Step Fitting Process

Using three data sets to get fitting for difference curves

7 Parameters

S Curve
Media Effect
Credit Availability
WALA
Burnout
Loan Size
CLTV

Refinance model v2

3 Step Fitting Process

Retain the 3 steps process in model v1

11 Parameters

S Curve
Media Effect
Credit Availability
WALA
Burnout
Loan size
CLTV(Normal / HARP)
HARP-Eligible
Refi-Elig %
HARPed %
TPO %

1 Constraint

Constraints for extrapolation

3 Step Fitting Process

Retain the 3 steps process in model v1.

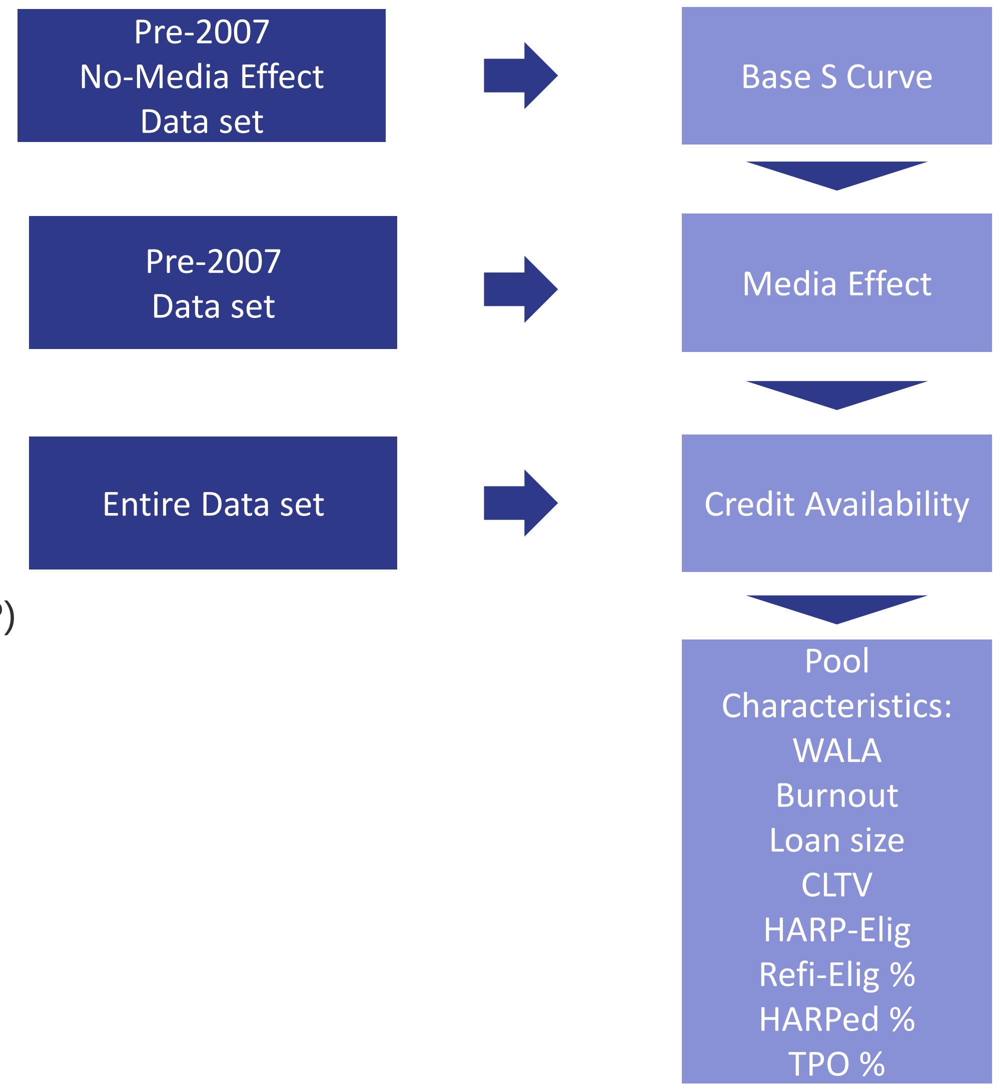
11 Parameters

S Curve
Media Effect
Credit Availability
WALA
Burnout

Loan size
CLTV(Normal / HARP)
HARP-Eligible
Refi-Elig %
HARPed %
TPO %

1 Constraint

Constraints for extrapolation



Question

- Our current incentive calculation only measures the refinance incentive for conventional 30-year borrowers to refinance into another 30-year conventional mortgage. If so, how we can capture that?

Problem

- Other than refinance into conventional programs, current conventional borrowers do have other choices, like refi into FHA loans.
- A rational borrower will choose the program that offers the largest incentive.
- Currently, as FHA doesn't have a risk-based MIP, borrowers with poorer credit profiles (higher CLTV/lower FICO) have greater incentive to refinance into FHA loans.

Solution

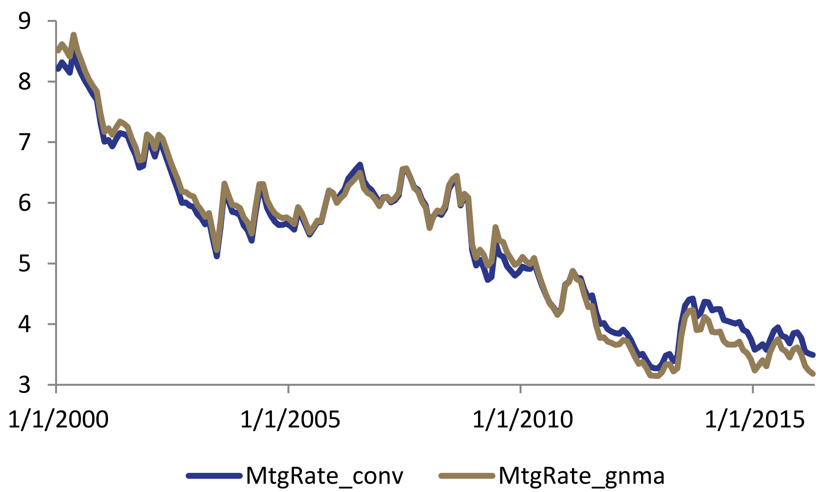
- We separately calculate the incentive for a conventional loan to respectively refi into another conventional loan and an FHA loan.
- The maximum of these two incentives corresponds to the "true" refinance incentive.

Incentive	**=**	**Current Loan Cost**	**-**	**New Loan Cost**
Conv. Incentive	=	(OrigNoteRate + PMI)	-	(PMMS + LLPA/4 + PMI)
FHA Incentive	=	(OrigNoteRate + PMI)	-	(FHA Mgt Rate + Upfront MIP/12 + Annual MIP)

Overall Incentive = Max (Conv. Incentive, FHA Incentive)

2 Major Differences

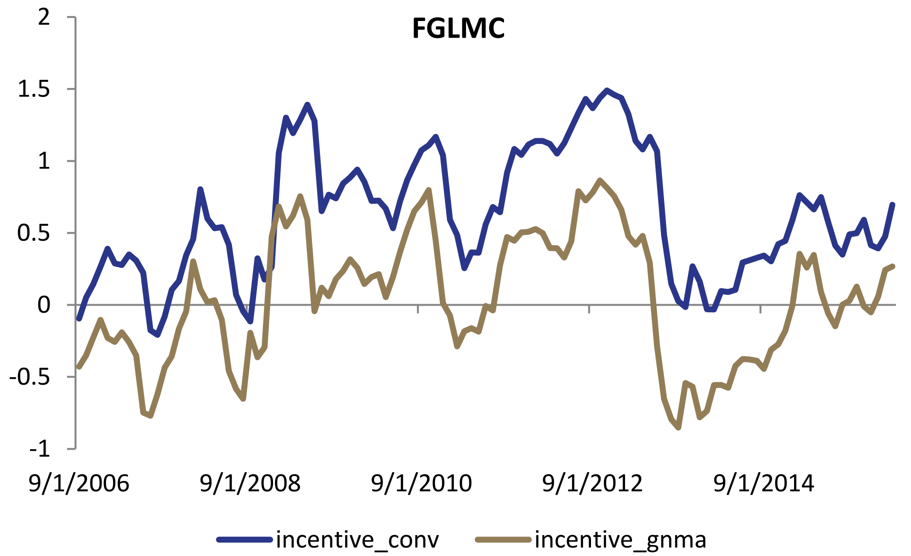
1. Mortgage Rate

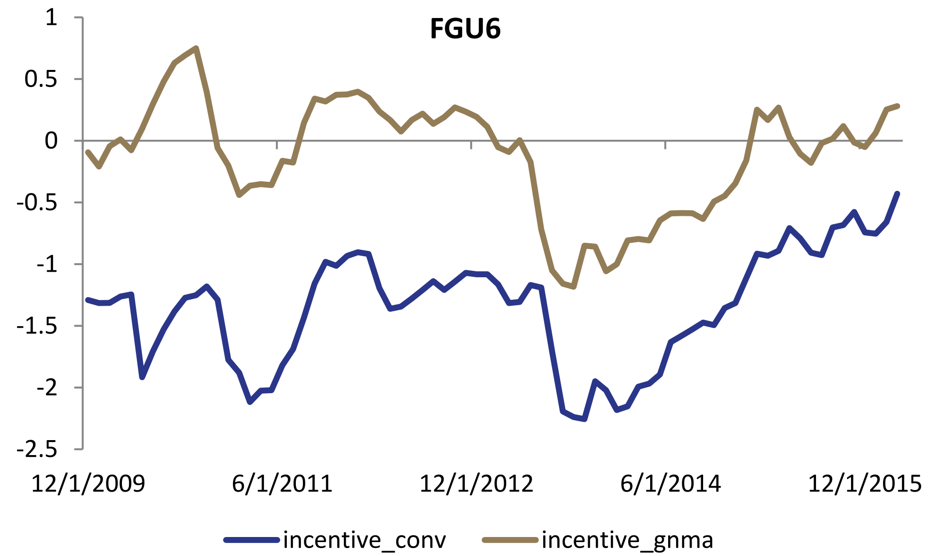


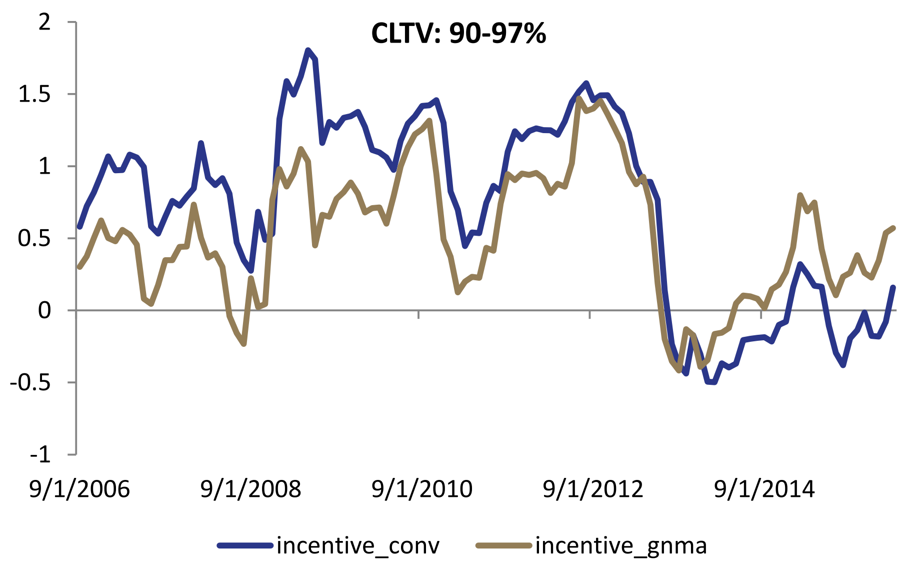
2. Risk-based Costs

LTV - Lower	LTV - Upper	760+	740-759	720-739	700-719	680-699	660-679	640-659	620-639
95.01	97	-.17	.03	.29	.62	.87	1.55	1.83	2.22
90.01	95	-.44	-.26	-.05	.21	.48	1.07	1.28	1.51
85.01	90	-.55	-.44	-.29	-.06	.13	.65	.83	1.00
0.01	85	-.66	-.65	-.55	-.39	-.28	.06	.21	.35

Conv-to-FHA Cost Difference (4/4/2016)







- For generic FGLMC pools, we see that the FHA incentive is lower than the conventional incentive
- FHA incentive increases with higher LTV. For 90-97% LTV, we see that FHA incentives are greater since 2013
- For U6 pools, FHA incentives are much greater.

As FHA doesn't have a risk based MIP, borrowers with poor profile(higher CLTV/lower FICO) has bigger incentive to refi into FHA loans

Question

- When we initially calculated burnout, we used the accumulated relative incentive. The relative incentive is defined as (OrigNoteRte / PMMS).
- This calculation needs to take LLPA and PMI into consideration.

Problem

- Need to re-calculate burnout on loan level

Solution

- We need to calculate relative incentive on loan level with LLPA and PMI
- Calculate a loan-level weighted average to calculate pool burnout.

Burnout in Model v1

- Pool level relative incentive = WAC / PMMS
- Special adjustments for when there's a gap between loan age and pool age.

Burnout in Model v2

- Weighted average of loan level relative incentive
- Loan level relative incentive
  = (OrigNoteRate + OrigPMI) / (PMMS + LLPA + PMI)
- Use pool-level WAC / PMMS to fill the gap between loan origination date and the first factor date

Step 1

$$Relative\ Incentive = \frac{OrigNoteRate + OrigPMI}{PMMS + LLPA + PMI}$$

Step 2

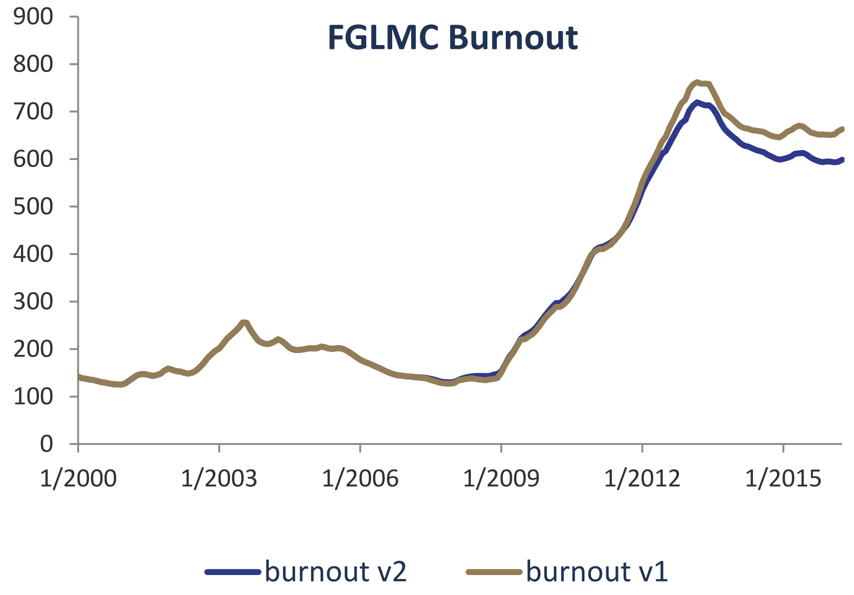
$$Log\ Incentive = Log\left(\frac{OrigNoteRate + OrigPMI}{PMMS + LLPA + PMI}\right)$$

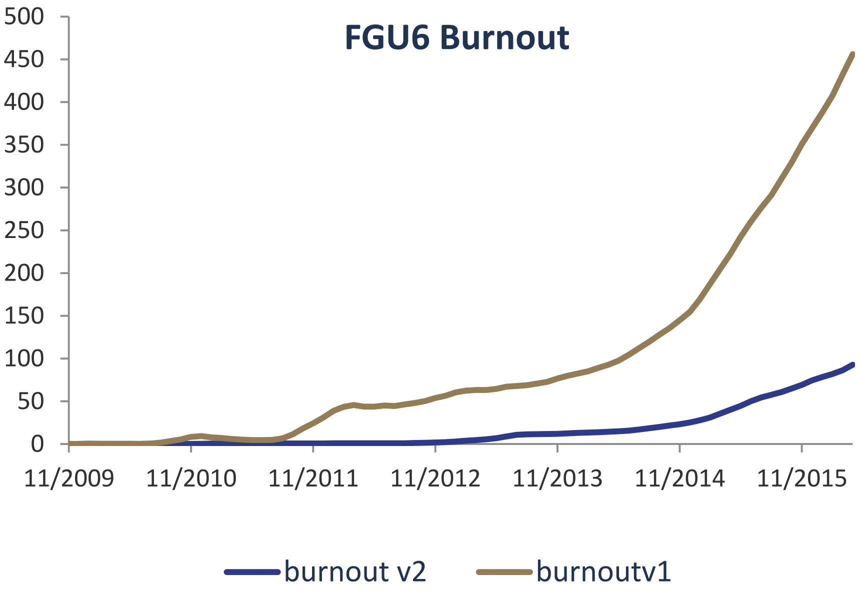
Step 3

$$Burnout = \sum_{0}^{t} \max\{Log\ Incentive, 0\}$$

Step 4

$$Adjusted\ Burnout = Burnout + Origination\ Adj.$$





- For conventional pools, the updates don't change aggregate burnout by much
- The revised calculation is significantly lower for high LTV pools.

Question

- The LLPA and PMI underwriting guidelines constrain the maximum CLTV and minimum FICO
- How do we measure the number of loans that are not eligible to refinance?

Problem

- Two practices we adopted in model v1 for ineligible mortgages are:
  - Assign the highest PMI / LLPA value in the underwriting matrix to these mortgages
  - Assign a very large number (like 999 bps) to the PMI/LLPA value to these mortgages
- However, neither of these two is a good solution. Using maximum values is misleading in that loans that are ineligible for refinancing will see an incentive if mortgage rates fall enough. Using very large numbers causes problems at the pool-level when we calculate a loan-level weighted-average refinance incentive.

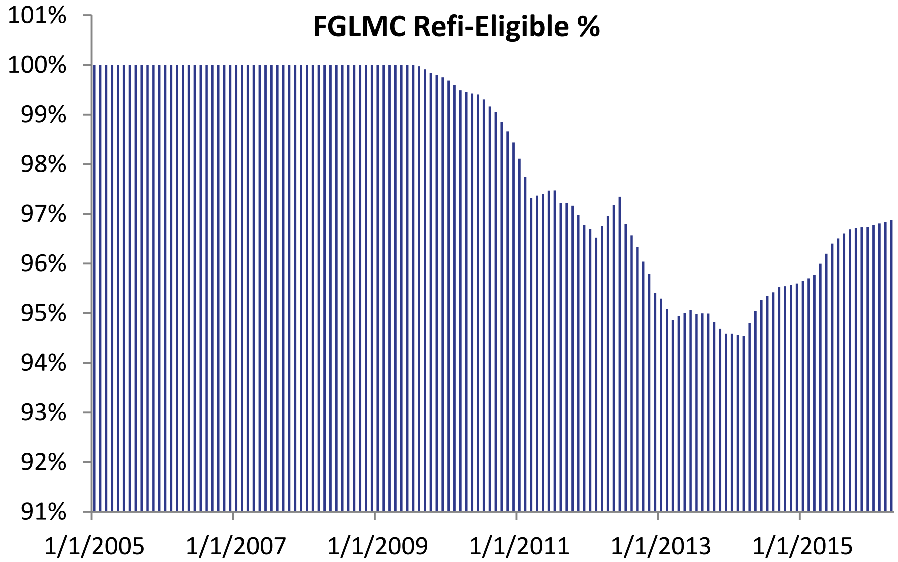
Solution

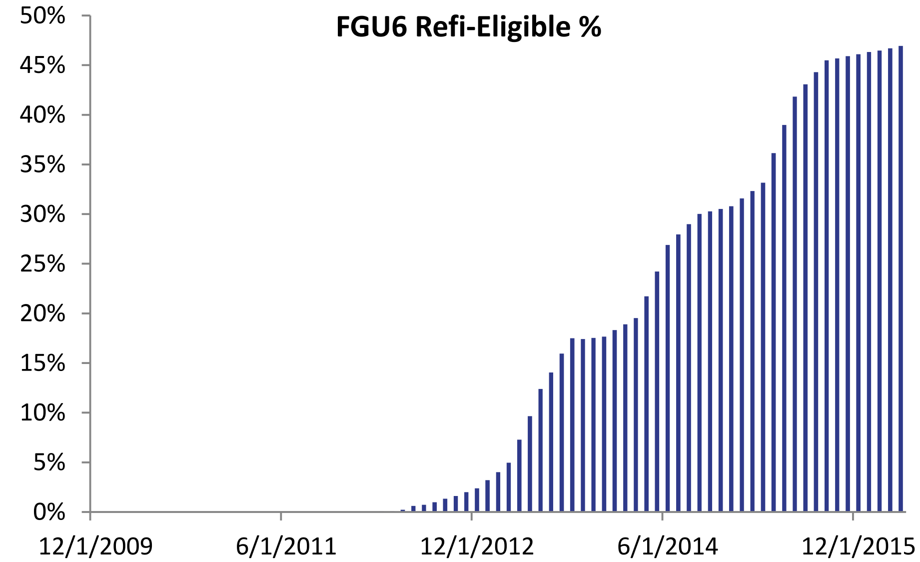
- Based on prevailing underwriting guidelines, we identify if each loan is eligible for refinancing or not.
- The refinancing test is “global” in that it should apply across different mortgage products that a conventional borrower could theoretically refinance into (i.e., the refinance eligibility test shouldn’t just apply for refinancing into another 30-year conventional mortgage)

FNMA Upfront Loan-Level Price Adjustments (AMDC of 0.25% & High LTV of 0.5% have been included)

	LTV								
Credit Score	**<= 60**	**60.01-70**	**70.01-75**	**75.01-80**	**80.01-85**	**85.01-90**	**90.01-95**	**95.01-97**	**>= 97.01**
> 740	0.00	0.25	0.25	0.50	0.25	0.25	0.25	0.75	Not Elig
720 - 739	0.00	0.25	0.50	0.75	0.50	0.50	0.50	1.00	Not Elig
700 - 719	0.00	0.50	1.00	1.25	1.00	1.00	1.00	1.50	Not Elig
680 - 699	0.00	0.50	1.25	1.75	1.50	1.25	1.25	1.50	Not Elig
660 - 679	0.00	1.00	2.25	2.75	2.75	2.25	2.25	2.25	Not Elig
640 - 659	0.50	1.25	2.75	3.00	3.25	2.75	2.75	2.75	Not Elig
620 - 639	0.50	1.50	3.00	3.00	3.25	3.25	3.25	3.50	Not Elig
< 620	Not Elig	Not Elig	Not Elig	Not Elig	Not Elig	Not Elig	Not Elig	Not Elig	Not Elig
Product Feature (Cumulative)									
Investment Property	2.125	2.125	2.125	3.375	4.125	Not Elig	Not Elig	Not Elig	Not Elig
2 Unit Property	1.00	1.00	1.00	1.00	1.00	Not Elig	Not Elig	Not Elig	Not Elig
Condominiums	0.00	0.00	0.00	0.75	0.75	0.75	0.75	0.75	Not Elig
Loan Size > 417K	0.25	0.25	0.25	0.25	0.25	Not Elig	Not Elig	Not Elig	Not Elig

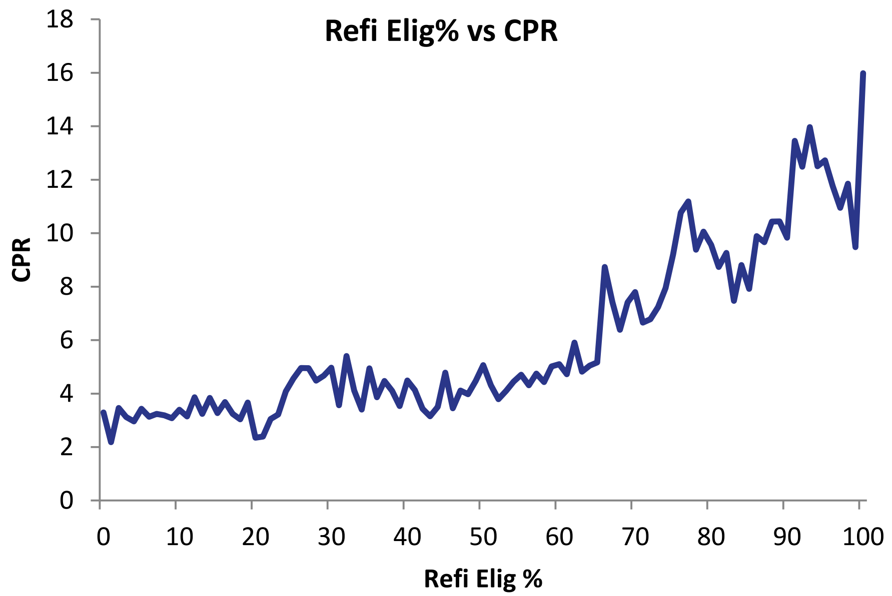
- Based on loan level information, we determine whether a borrower is eligible to refinance under current underwriting guidelines. This flag is **time-dependent** in that underwriting guidelines vary over time.
- We then calculate the loan-level weighted average for each pool.
- Pools originated before 2009-06 are 100% Refinance Eligible after 2009-06 because of the HARP program
- Note that treating a pool like a loan and calculating a refi-eligible flag is not practical. For example, a pool with an LTV of 96% would technically be 100% Refinance Eligible even though there may be several loans with LTVs > 97%.





- Refi-eligible percentage for conventional pools experienced a 5% decrease from 2009 to 2014.

- When U6 pools get originated, all of them have OLTV greater than 97%. So in that case 0% of the balance is eligible to refi.
- As loan ages increase, CLTV typically decreases. Around 50% of these loans are eligible to refi by now.



Question

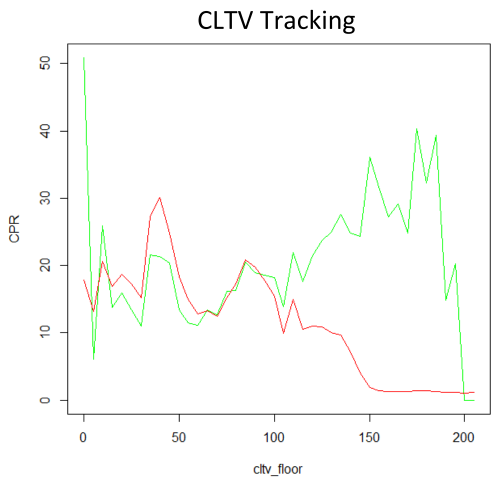
- In general it is harder for mortgages to refinance when their CLTVs go up. However, this pattern doesn't apply to HARP-eligible mortgages.
- So, how can we capture the difference between normal mortgages and HARP eligible mortgages?

Problem

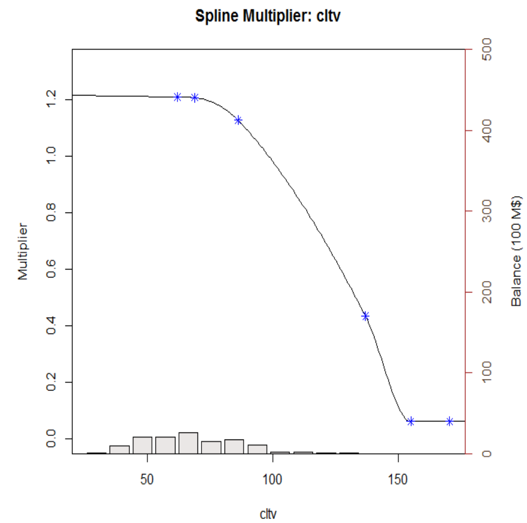
- Borrowers with very high CLTVs have difficultly refinancing but HARP eligible borrowers appear to behave differently -- they refinance faster at higher CLTVs.
- HARP eligible is a binary factor for the mode; linking it directly as a multiplier rescales the normal CLTV curve. However, given the opposing pattern of prepayment behavior as a function of CLTV for HARP and non-HARP eligible borrowers, this rescaling leads to significant distortions.

Solution

- We use the HARP eligibility as a switch and build two CLTV curves.
  - CLTV Curve 1: calibrate and apply on normal mortgages
  - CLTV Curve 2: calibrate and apply on HARP eligible mortgages
- When a pool is HARP eligible, choose CLTV curve 2, otherwise use CLTV curve 1.

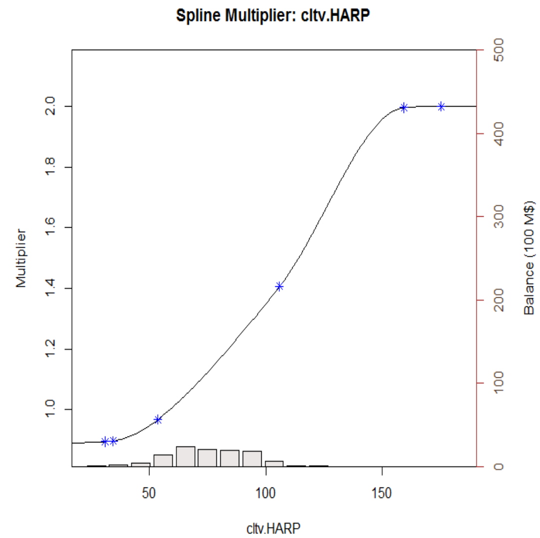


Not eligible for HARP

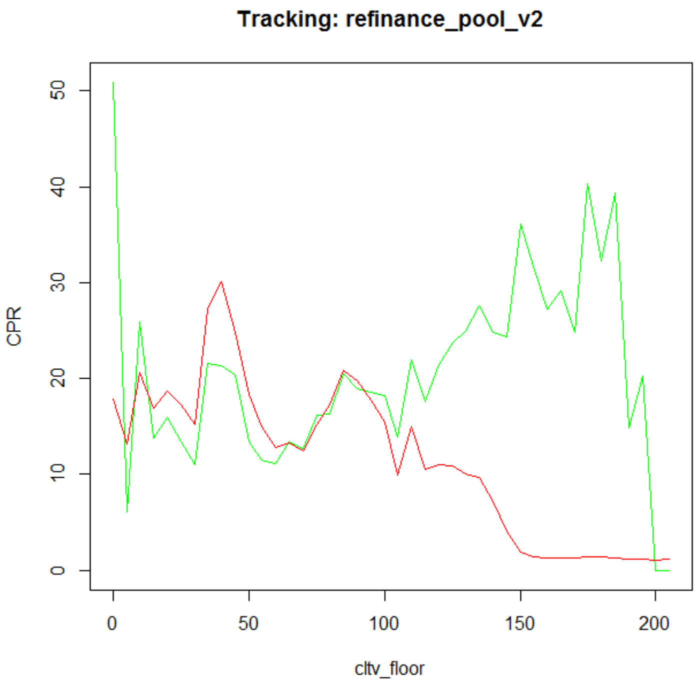


Check origination date:
If originated before 2009-06

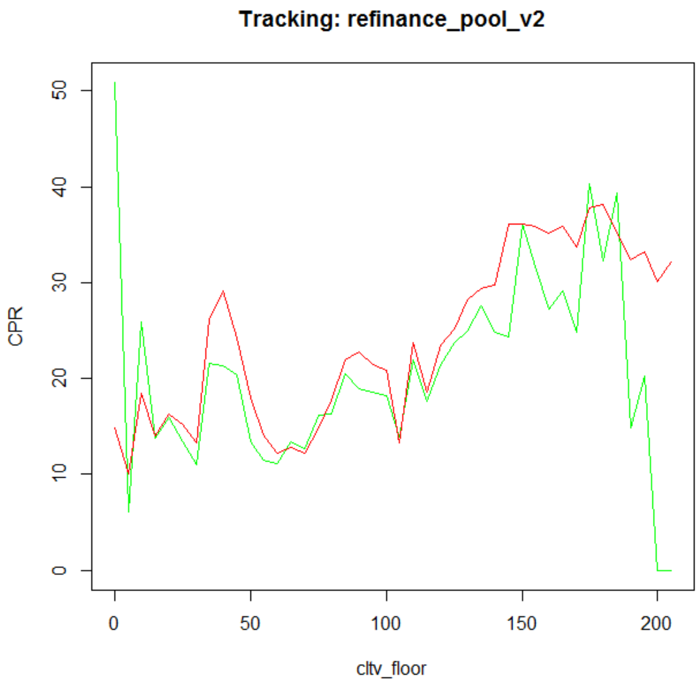
Eligible for HARP



- Notice the difference in behavior by the two groups.



Using One CLTV Curve

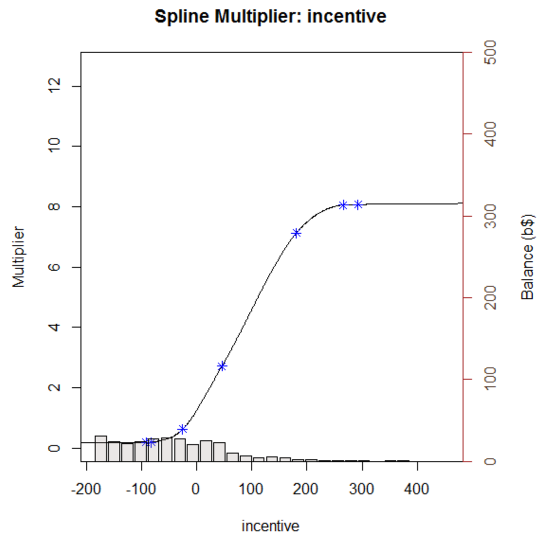


Using Two CLTV Curves
And HARP Eligible Factor

S-Curve	The baseline curve is built based on the pre-2007 no-media-effect period. Incentive is the max of Conventional and FHA Incentive.
Media Effect	Using Refinanceable Universe as Media Effect. Curve is calibrated based on pre-2007 data.
Credit Availability	HPA 2yr MA. Higher levels of HPA are correlated with greater mortgage credit availability.
WALA	Short seasoning ramp for refinancings
Burnout	Cumulative incentive in a loan's life time.

Loan Size	Large size loans will refi faster than loans with smaller balances.
CLTV	Borrowers with higher LTVs are less likely to refinance. However, the HARP program made it possible for even very high LTV loans to refinance
Refi Elig %	The percentage of a pool's balance that is eligible to refinance.
HARPed %	HARPed loans are more difficult to refi compared to Purchase loans originated over the same time period
Origination Channel	Third party-originated loans refinance faster

- v2 of the Refinance model includes the 10 prediction parameters listed above.
- Sequentially fitting model with the 10 parameters, we can create a cubic spline curve for each one of these inputs.
- The parameter ordering is based on a single parameter regression calculated R-square value.
- Parameters that have earlier order will have more significant impact on our model output.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Multiplier
Balance (b$)
incentive
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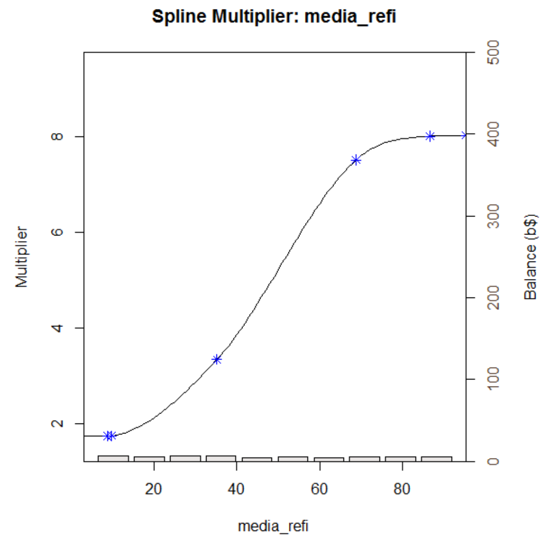
-100
0
100
200
300
400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Multiplier
Balance (b$)
media_refi
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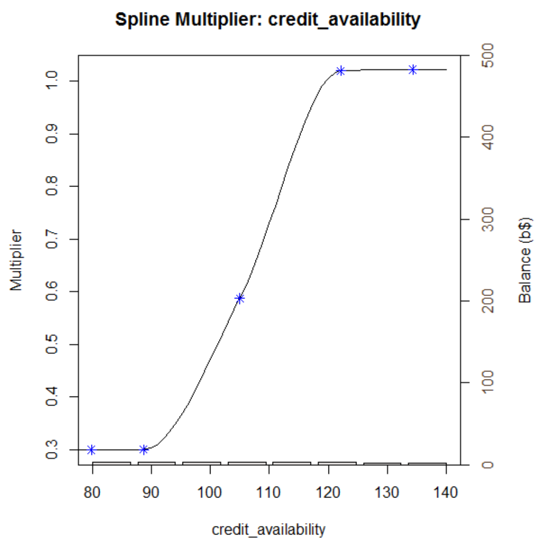
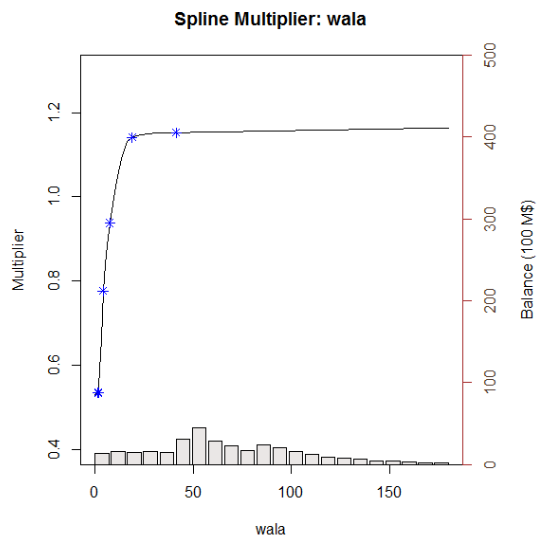
40
60
80


Spline Multiplier: credit_availability
Multiplier
Balance (b$)
credit_availability
80
90
100
110
120
130
140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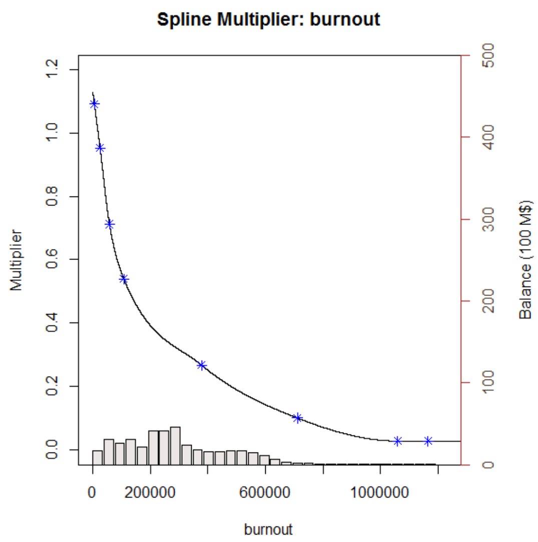
Multiplier
Balance (100 M$)
wala
0
50
100
150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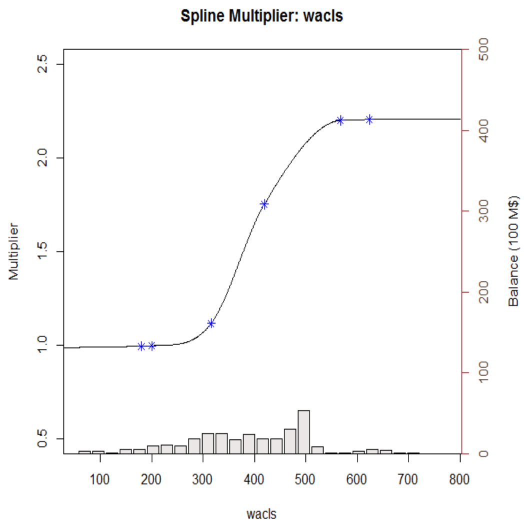
Multiplier
Balance (100 M$)
burnout
0
200000
600000
1000000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Balance (100 M$)
wacls
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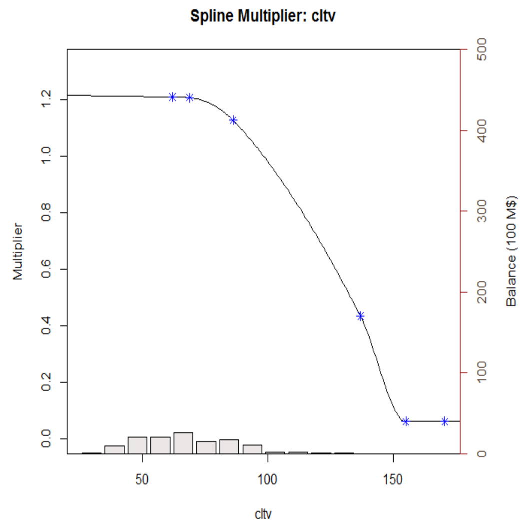
0.5 1.0 1.5 2.0 2.5
0 100 200 300 400 500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Multiplier
Balance (100 M$)
cltv
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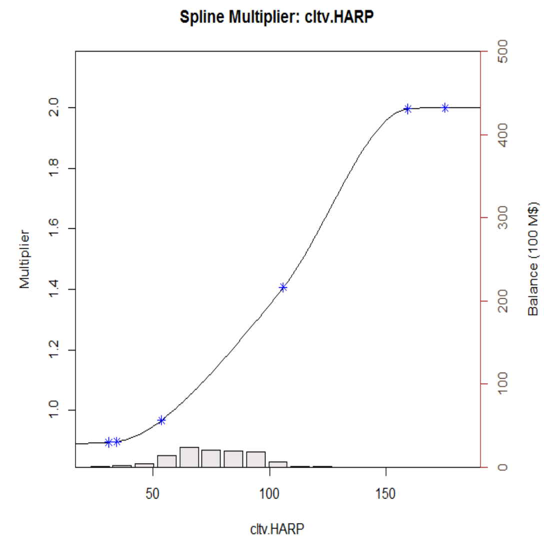
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Multiplier
Balance (100 M$)
cltv.HARP
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1.0 1.2 1.4 1.6 1.8 2.0
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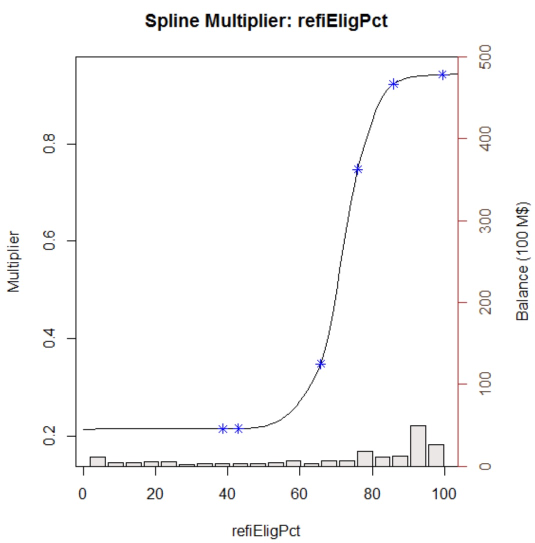
Spline Multiplier: refiEligPct
Multiplier
Balance (100 M$)
refiEligPct
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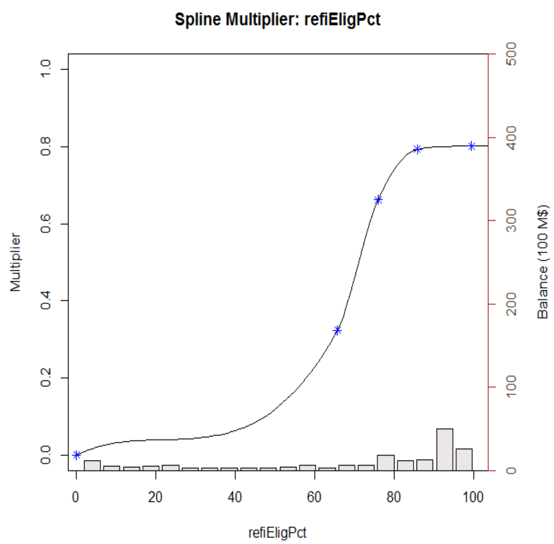
0.2 0.4 0.6 0.8
0 100 200 300 400 500


Spline Multiplier: refiEligPct
Multiplier
Balance (100 M$)
refiEligPct
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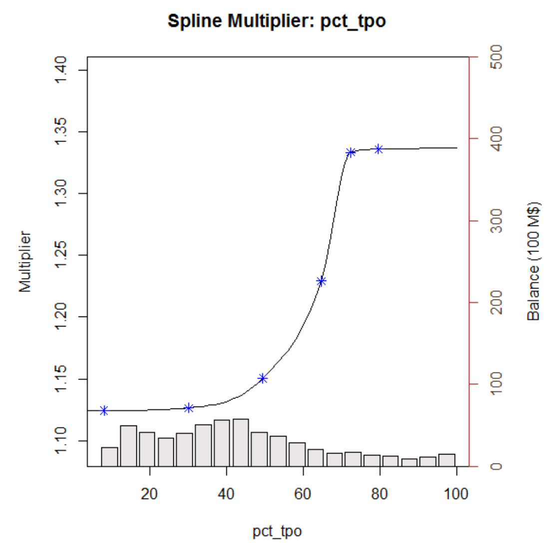
0 100 200 300 400 500


Spline Multiplier: pct_tpo
Multiplier
Balance (100 M$)
pct_tpo
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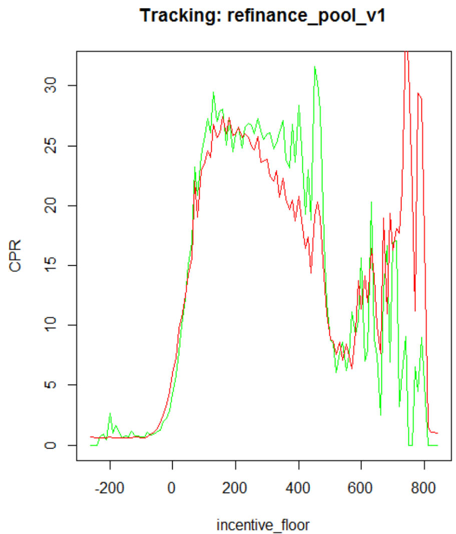
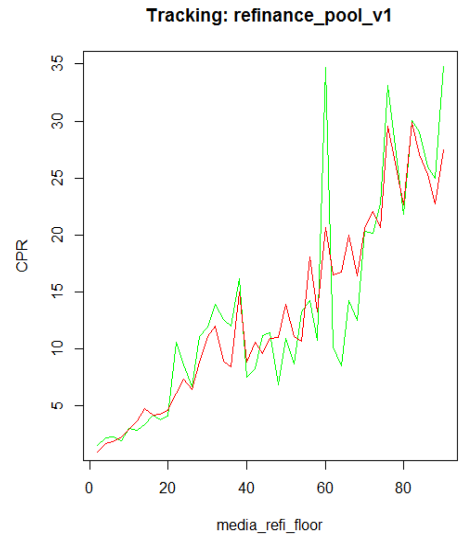
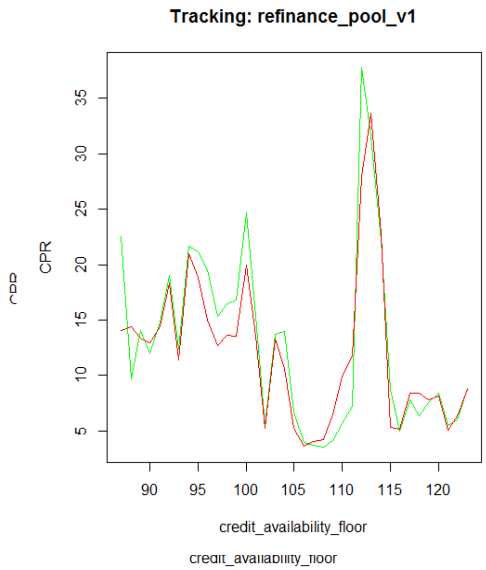
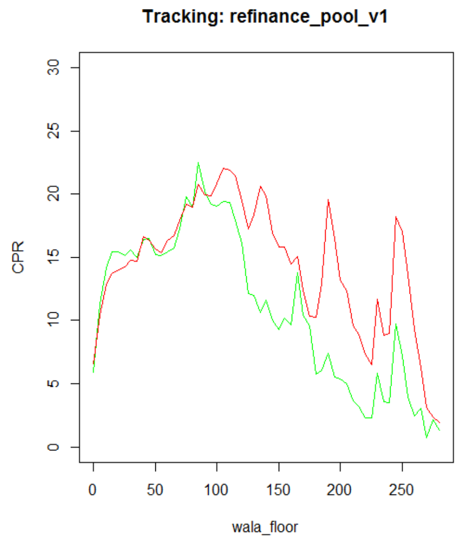
1.10 1.15 1.20 1.25 1.30 1.35 1.40
0 100 200 300 400 500

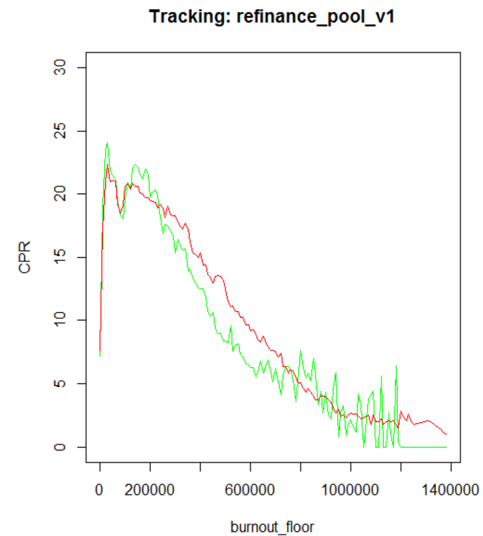
v2 Refinance Model Details
3. Model Multiplier Tracking



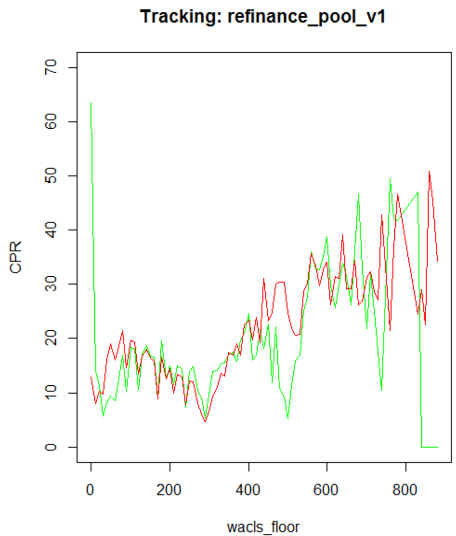


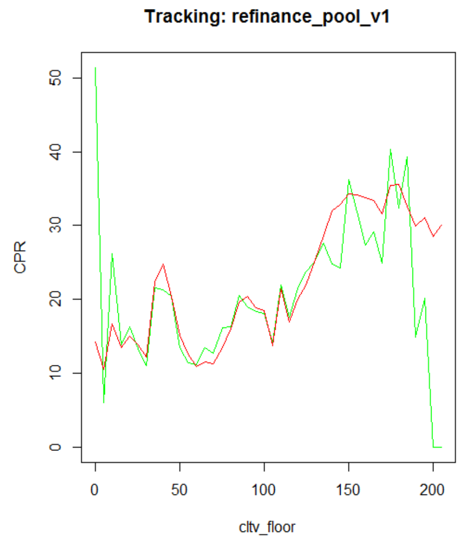


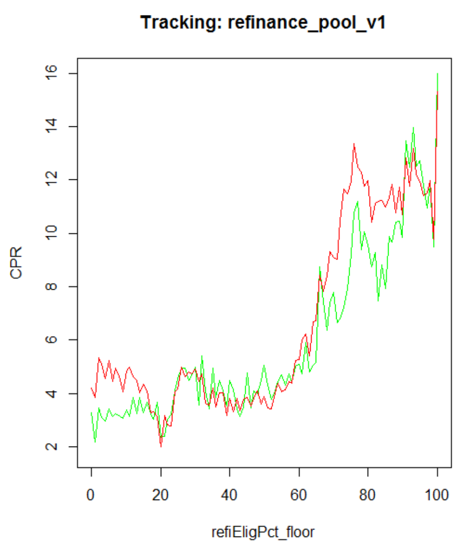


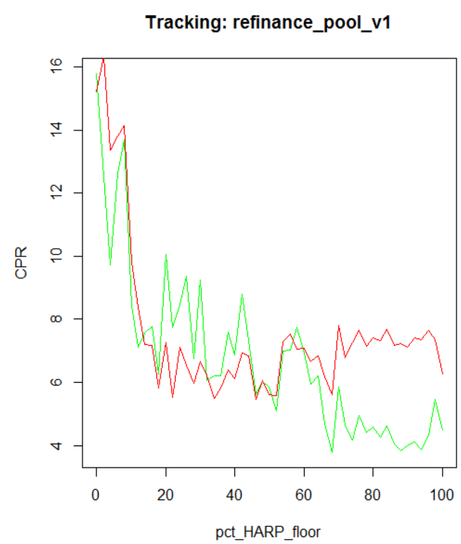


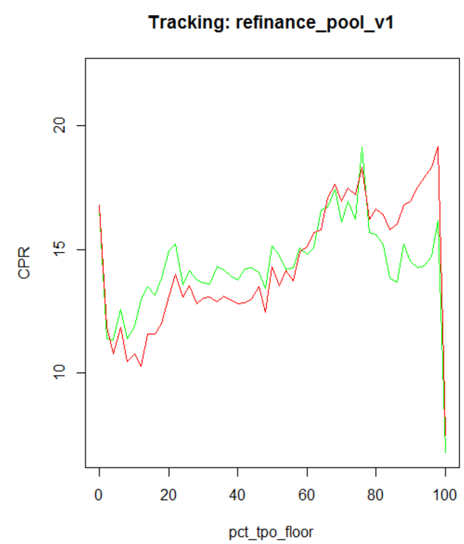
Actual
Model



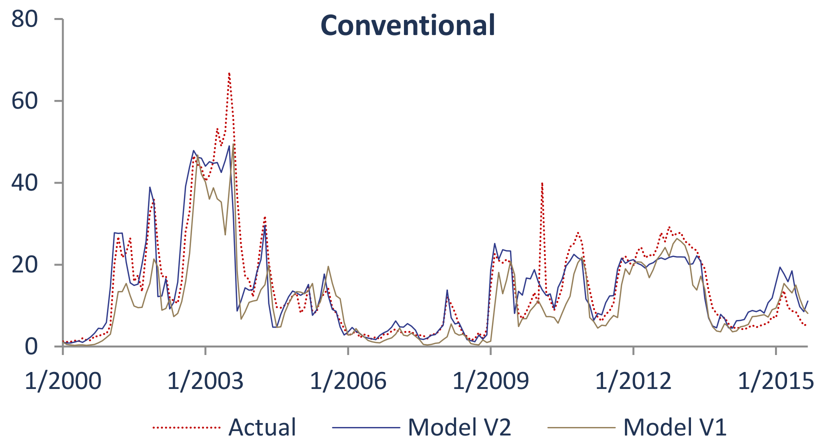




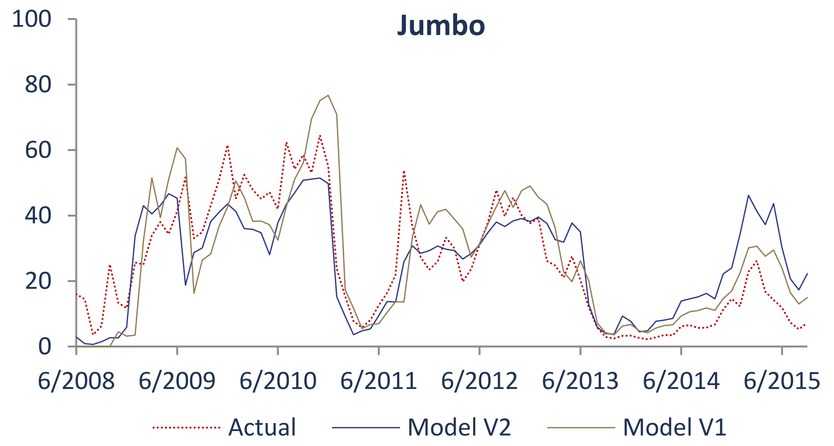




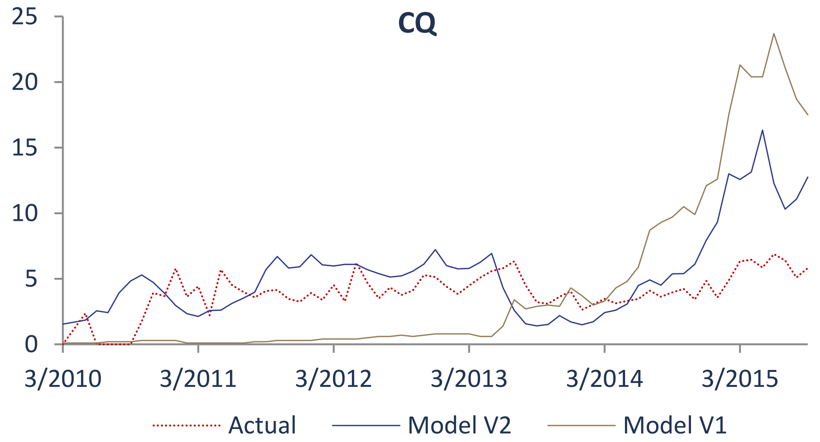
Actual
Model



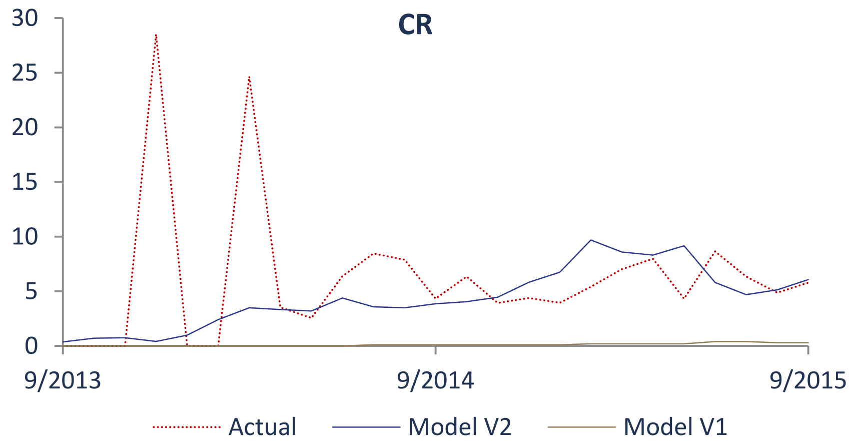
Conventional pool tracking shows v2 has smaller tracking error.



Jumbo pools tracking for post 2014 looks faster for v2 than v1



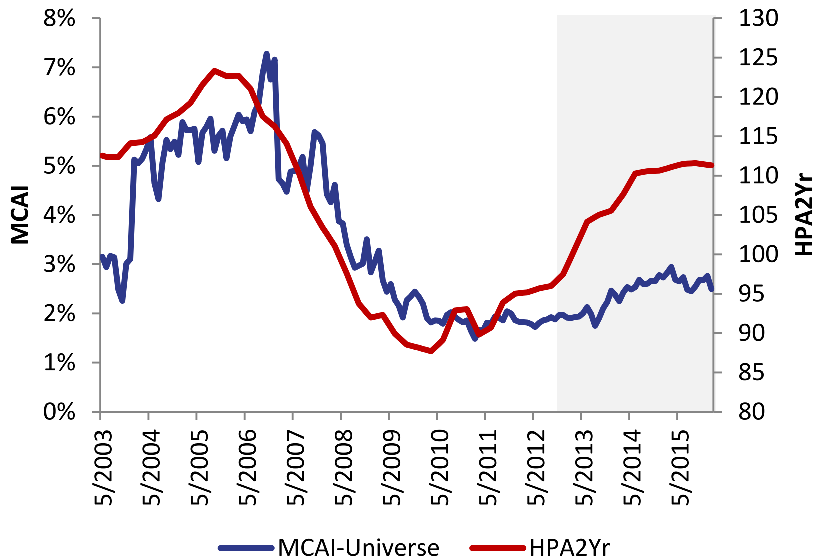
CQ pool tracking is more reliable now, as we fixed the incentive calculation for HARPed loans.

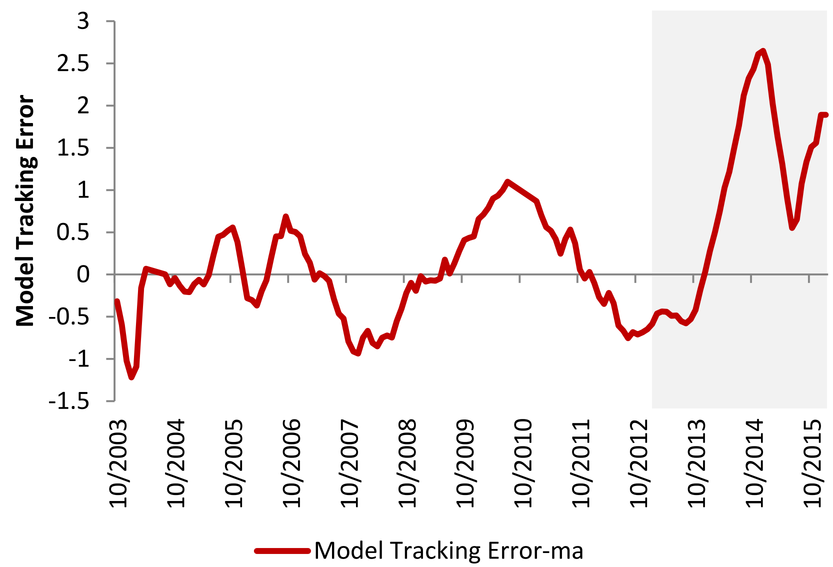


Similar as CQs. V2 model shows better performance for CRs

	Min	Max
baselineConstant	1.86	
incentive	0.19	8.08
media	1.75	8.02
creditAvailability	0.30	1.02
burnout	0.03	1.09
wacls	0.99	2.33
cltv	1.02	1.94
refiElig	0.21	0.94
pct_HARP	0.45	0.87
pct_tpo	1.12	1.34
Multiplier	0.00	358.71

Dispersion Between MCAI and HPA2Yr

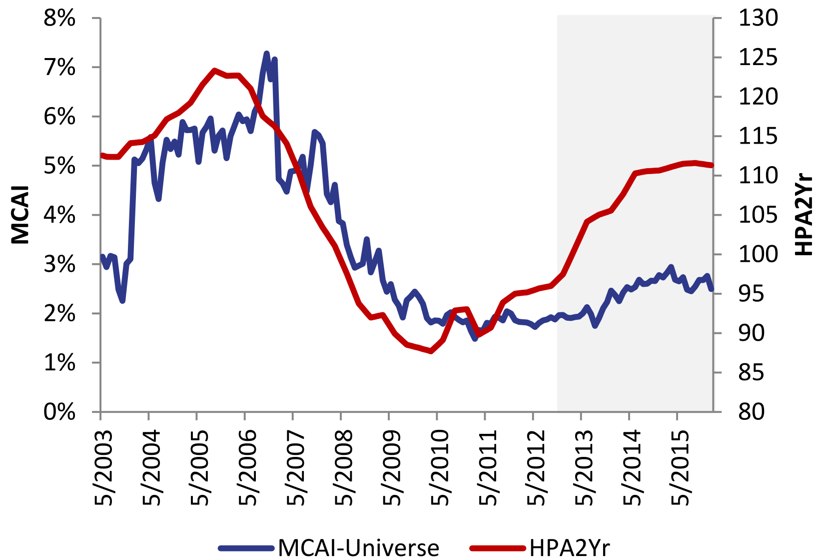


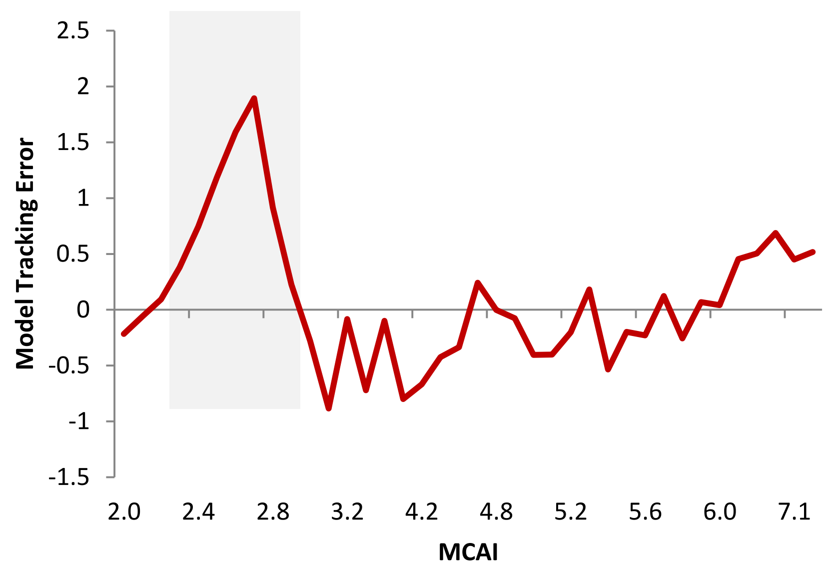


In our current model, we use HPA two-year moving average as a measure of market credit availability. However, comparing the Mortgage Credit Availability Index and HPA2Yr curves, we can see a dispersion between them after 2013. At the same time, our model tracking error goes up.

We might want to change the credit availability measure in our future version of model. MCAI would be a choice. In that case, since the MCAI is calculated by the origination information of all kinds of loan products(including non-agency products), we need to actively maintain the market share of different products.

Dispersion Between MCAI and HPA2Yr





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- **Burnout**: Incorporate FHA Incentive in measuring incentive
- **Burnout**: Backfill history for seasoned loans using LLPA and PMI (recall that OLTV is available at the loan-level)
- **Refi Eligibility**: Ensure that this is a global flag – i.e., it measures the eligibility of a loan to refinance into *any* mortgage program
- **Refi Eligibility**: Are we capturing the ability of high-balance loans to refinance into an FHA Jumbo mortgage?
- **MCAI**: Refit MCAI multiplier using MBA MCAI (2-year moving average). Also try Cello MCAI.